ASSESSMENT 3: PROJECT BRIEF

Subject Code and Title	CAI104
Assessment	Assessment 3: Project
Individual/Group	Individual
Length	Project and supporting document (2000 words) (10% +-)
Submission	12-Week Duration: Due by 11:55pm AEST/AEDT, Wednesday of Week 12 6-Week Duration: Due by 11:55pm AEST/AEDT, Wednesday of Week 6
Weighting	40%
Total Marks	100 marks

Task Summary

In this assessment, you are to program an AI agent to solve a real-world, challenging case study. This assessment is done individually and you are to submit programs and supporting documents. Please refer to the Task Instructions for details on how to complete this task. This assessment is intended to test:

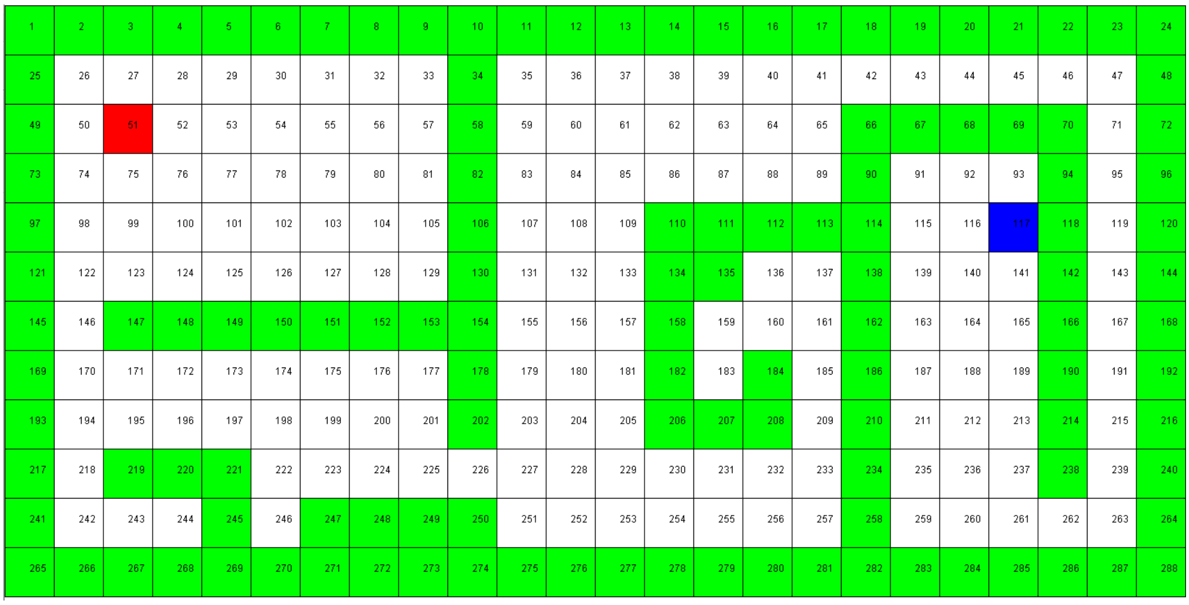
- Your understanding of the theories covered in Module 1 to 12.
- Your ability to formulate and frame a simplified real-world problem for an AI problem solving technique.
- Your ability to choose a suitable AI technique for the problem.
- Your ability to develop an AI problem solving technique in a modern programming language.
- Your ability to provide a document to discuss the potential applications and ethics of the AI solution.

Context

You are to create a robot path planner that is able to find an optimal path to navigate an environment and reach a target. By completing this assessment, you will show your skills on leveraging the best AI methods to solve a simplified real-world problem.

The maze can be seen in the image below. It can be seen that there are 12 rows and 24 columns, meaning there is a total of 288 blocks on the map. There are four different types of blocks in this map as follows:

- Green: wall
- White: space (void)
- Red: initial position of the robot
- Blue: the target



You can easily represent the entire maze as a 2D array with 0s and 1s:

```
{1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1},
{1,0,0,0,0,0,0,0,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,1},
{1,0,-1,0,0,0,0,0,0,1,0,0,0,0,0,0,0,1,1,1,1,1,0,1},
{1,0,0,0,0,0,0,0,0,1,0,0,0,0,0,0,0,1,0,0,0,1,0,1},
{1,0,0,0,0,0,0,0,0,1,0,0,0,1,1,1,1,1,0,0,9,1,0,1},
{1,0,0,0,0,0,0,0,0,1,0,0,0,1,1,0,0,1,0,0,0,1,0,1},
{1,0,1,1,1,1,1,1,1,1,0,0,0,1,0,0,0,1,0,0,0,1,0,1},
{1,0,0,0,0,0,0,0,0,1,0,0,0,1,0,1,0,1,0,0,0,1,0,1},
{1,0,0,0,0,0,0,0,0,1,0,0,0,1,1,1,0,1,0,0,0,1,0,1},
{1,0,1,1,1,0,0,0,0,0,0,0,0,0,0,0,0,1,0,0,0,1,0,1},
{1,0,0,0,1,0,1,1,1,1,0,0,0,0,0,0,0,1,0,0,0,0,0,1},
{1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1},
```

In this array, we use the following numbers to represent different types of blocks:

- 1: wall
- 0: space (void)
- -1: initial position of the robot
- 9: the target

In this assessment, you need to design and implement a robot path planner. Note your learning facilitator will give you the type and name of the search or optimization algorithms that you have to implement. The learning facilitator might ask you to work on a randomly generated maze as well. You do not have to draw a maze like the picture above. You can simply mark the shortest path obtained by your algorithm using for instance 2 (or any other numbers except 1, 0, -1, and 9) in the 2D array. If you draw the maze, however, it will be a lot easier to visualise the path and make sure that it is the closest path to the target when you are testing your program.

By implementation, we mean to write the steps that should be taken to find the shortest path from the initial position to the target as a pseudocode.

After implementation, write a reflective report detailing the experience of the development process. The report should be 2000 words (10%+-) in length and include the following sections:

- Overview
- What went right
- What went wrong
- What you are not sure about
- Conclusion

Note that your pseudocode is not be included in the word count.

Task Instructions

- Appropriate, effective and correct usage of pseudocode to design and develop and algorithm
- Effective use of search and or optimization algorithms in AI.
- Good selection of search and/or optimization algorithms.

The pseudocode that you will be submitting:

1. Should be clear and detailed
2. Should be structured and written with the best practices.
3. There should be enough number of comments to show your understanding of the program.
4. Flowcharts can be used to complement the pseudocode

When you submit the electronic version of your project make sure to use the following names:

- Name the source code folder as: Source – Student Name
- Name the solution as: YourGameName.sln

Submission Instructions

Please submit an MS Word or a PDF file including your pseudocode and reflective report.

Assessment Rubric

Assessment Attributes	Fail (Yet to achieve minimum standard) 0-49%	Pass (Functional) 50-64%	Credit (Proficient) 65-74%	Distinction (Advanced) 75-84%	High Distinction (Exceptional) 85-100%
Work demonstrates the knowledge and understanding of the best knowledge representation methods for the case study considered in the assessment 40%	Little or no knowledge of the best knowledge representation methods for the case study considered in the assessment. A state space tree or other standard AI representation methods are not used.	Acceptable but further work is required to show the knowledge of the best knowledge representation methods for the case study considered in the assessment. A state space tree or other standard AI representation methods are used but include errors and flaws.	Good level of knowledge about the best knowledge representation methods for the case study considered in the assessment. A state space tree or other standard AI representation methods are used but not in an efficient manner for the problem.	Excellent but not thorough knowledge about the best knowledge representation methods for the case study considered in the assessment. A state space tree or other standard AI representation methods are used but it is not robust and error free for different mazes.	Excellent and thorough understanding of the best knowledge representation methods for the case study considered in the assessment
Work demonstrates the knowledge and understanding of the search algorithm for the case study considered in the assessment 40%	Little or no knowledge of the search algorithms for the case study considered in the assessment. The search method is attempted but not implemented correctly.	Acceptable but further work is required to show the knowledge of the search algorithms for the case study considered in the assessment. The search method is implemented but includes errors and flaws.	Good level of knowledge about the search algorithms for the case study considered in the assessment. The search method is implemented but not in the most efficient manner.	Excellent but not thorough knowledge about the search algorithms for the case study considered in the assessment. The search method is efficient but it is not robust and error free for different mazes.	Excellent and thorough understanding of the search algorithms for the case study considered in the assessment. The search method is highly efficient, robust, and error free.
The reflective essay demonstrates the knowledge and understanding of the whole process of problem solving using the best AI problem solving methods and practices	The reflective essay includes no or little sections and concepts required. There is no or little elaborations or justifications.	The reflective essay includes some of the sections and concepts required. There is little elaborations or justifications to demonstrate the knowledge and understanding of the whole process of problem solving using the best AI problem	The reflective essay includes all the sections and concepts required. Elaborations and justifications are not discussed well to show the mastery of the AI technique used to solve the problem.	The reflective essay includes all the sections and concepts required. Elaborations and justifications are not thorough and in-depth to demonstrate the knowledge and understanding of the whole process of problem	The reflective essay includes all the sections and concepts required. Elaborations and justifications are thorough and show the mastery of the process of solving a simplified real-world problem using an AI

20%		solving methods and practices.		solving using the best AI problem solving methods and practices.	technique.

The following Subject Learning Outcomes are addressed in this assessment	
SLO a)	Demonstrate an understanding of the field of artificial intelligence including its origins, fundamental concepts, limitations and the problems that it tries to solve.
SLO b)	Explain basic concepts, methods, theories and application for search.
SLO c)	Interpret and formulate knowledge representations in the form of logic expressions.
SLO d)	Describe knowledge systems, the use of AI based problem solving methods, forms of knowledge representation, and model-based reasoning.
SLO e)	Explain basic concepts, methods, theories and application for learning.
SLO f)	Discuss the philosophical foundations and ethics of AI.